

Relational Model

Core Entities:

Patient (patient_id, first_name, last_name, date_of_birth, gender, phone, email, address, blood_type, allergies, #history_id)

Doctor (doctor_id, first_name, last_name, license_number, phone, email, hire_date, #specialty_id, #department_id)

Nurse (nurse_id, first_name, last_name, phone, hire_date, shift_type, #department_id)

Surgery (surgery_id, surgery_type, surgery_date, duration_min, outcome #room_id)

Doctor_Surgery (#doctor_id, #surgery_id)

Patient_Surgery(#patient_id, #surgery_id)

Medical Records:

VitalSigns(vital_id, heart_rate, systolic_bp, diastolic_bp, temperature, respiratory_rate, weight_kg, height_cm, pain_scale, #nurse_id, #consultation_id)

ClinicalHistory(history_id, event_type, condition_name, event_date, end_date, outcome, notes, #patient_id, #doctor_id)

Appointment (appointment_id, appt_date, appt_time, status, reason, #patient_id, #doctor_id, #staff_id)

Speciality (specialty_id, specialty_name, description)

Consultation (consultation_id, date, notes, #appointment_id, #room_id, #doctor_id)

Diagnosis (diagnosis_id, icd_code, description, severity, is_chronic, #consultation_id)

LabTest(test_id, test_name, ordered_date, result_date, result_value, status, #consultation_id)

Prescriptions:

Prescription (presc_id, issue_date, expiry_date, #consultation_id)

PrescriptionLine(line_id, dosage, frequency, duration_days, instructions, #presc_id, #med_id)

Medication (med_id, med_name, generic_name, dosage_form, unit, stock_qty)

Medication_Supplier(#med_id, #supplier_id)

Supplier (supplier_id, supplier_name, contact_person, phone, email, address)

Hospital Resources:

Hospitalisation(hosp_id, admit_date, discharge_date, reason, #patient_id, #room_id)

Room (room_id, room_number, room_type, capacity, is_available, # department_id)

Department (department_id, dept_name, floor, phone_ext)

Staff (staff_id, first_name, last_name, role, phone, email, hire_date, salary, contract_type, #department_id)

Equipment (equip_id, equip_name, purchase_date, status, last_service_date, #department_id)

Billing & Insurance:

Invoice (invoice_id, issue_date, total_amount, status, due_date, # patient_id, #staff_id)

Payment (payment_id, amount_paid, payment_date, method, # invoice_id)

Insurance (insurance_id, provider_name, policy_number, coverage_pct, expiry_date, #patient_id)

Certification:

MedicalCertification(cert_id, issue_date, valid_days, purpose, #consultation_id)